

“Very Likely” Question Bank

180 VCAA-aligned practice questions across the 18 highest-probability key knowledge points for 2026. Ten VCAA-style questions for each of the 18 highest-probability key knowledge points, tagged by action word, exam section and marks, with indicative answers for marking.

VCAA VCE Applied Computing — Software Development (2025–2028 Study Design)

18 KKs × 10 questions = 180 items

Sections A / B / C

Compiled 29 June 2026

How the tiers were derived: the “Very likely” set is the Tier 4 layer of the Digital Vector exam tracker — built from the predicted 2026 focus areas plus the “tested until mastered” rule applied to where the 2025 cohort lost the most marks. This is a revision-prioritisation aid and Digital Vector’s projection, **not** a VCAA forecast. Always check wording against the official VCAA study design and examination reports.

SECTION A

Multiple choice · 1 mark

SECTION B

Short answer (20 marks total)

SECTION C

Case-study short & extended (60 marks)

ACTION WORDS

State · Describe · Explain · Compare · Discuss · Justify...

THE 18 “VERY LIKELY” KEY KNOWLEDGE POINTS

U301·KK01	Emerging trends in programming using AI
U301·KK03	Design tools for representing modules — IPO charts, pseudocode, object descriptions
U301·KK06	Characteristics of data sources — files, CSV and XML
U301·KK07	Principles of object-oriented programming
U301·KK14	Debugging and testing techniques
U302·KK03	Project management — Gantt charts, dependencies, critical path, milestones
U302·KK04	Techniques for collecting data
U302·KK08	Depicting relationships — context diagrams, data flow diagrams, use case diagrams
U302·KK10	Key legal requirements — privacy and copyright
U302·KK15	Characteristics of user experiences (UX)
U302·KK16	Design principles influencing appearance and functionality
U401·KK04	Innovative approaches to software development
U402·KK02	Advantages and disadvantages of developing software in-house vs externally
U402·KK03	Vulnerabilities within insecure development environments (incl. combined dev/test/production)
U402·KK04	Security controls to protect development practices and data
U402·KK05	Threat modelling principles
U402·KK07	Key legislation affecting organisations and data
U402·KK08	Ethical issues arising when developing software

U301·KK01

UNIT 3 - OUTCOME 1 — PROGRAMMING

▲ VERY LIKELY

Emerging trends in programming using AI

2026 signal: AI ethics & emerging trends named as a leading 2026 emphasis.

Q01

IDENTIFY

SECTION A · MULTIPLE CHOICE

1 mark

A developer uses an AI coding assistant to generate a function. The MOST significant reliability concern is that the generated code may

- A. take longer to execute than hand-written code.
- B. contain logic errors or insecure patterns the developer does not detect.
- C. use a different naming convention to the rest of the project.
- D. require an internet connection to run.

INDICATIVE ANSWER / MARKING NOTE

B. AI-generated code can embed subtle logic/security flaws; over-trust without review is the key reliability risk (A/C/D are minor or not reliability).

Q02

DEFINE

SECTION B · SHORT ANSWER

1 mark

Define the term 'large language model' in the context of AI-assisted programming.

INDICATIVE ANSWER / MARKING NOTE

A model trained on large volumes of text/code that predicts likely next tokens to generate or complete code and explanations.

Q03

OUTLINE

SECTION B · SHORT ANSWER

2 marks

Outline two ways AI tools are changing how programmers develop software solutions.

INDICATIVE ANSWER / MARKING NOTE

Any 2 × 1: code generation/autocompletion; automated debugging or test generation; natural-language to code; rapid prototyping; documentation generation.

Q04

DESCRIBE

SECTION B · SHORT ANSWER

2 marks

Describe one benefit and one risk of using an AI assistant to generate code for a commercial solution.

INDICATIVE ANSWER / MARKING NOTE

Benefit (1): speed/productivity, boilerplate reduction, learning support. Risk (1): incorrect/insecure code, IP/licensing issues, over-reliance reducing developer understanding.

Q05

EXPLAIN

SECTION B · SHORT ANSWER

3 marks

A student submits AI-generated code without testing it. Explain why this is poor development practice, referring to reliability and accountability.

INDICATIVE ANSWER / MARKING NOTE

Reliability (1-2): AI output is probabilistic and may be wrong/insecure; untested code may fail on edge cases. Accountability (1): the developer remains responsible for correctness; lack of testing breaches professional practice.

Q06**DISCUSS****SECTION B · SHORT ANSWER****4 marks**

Discuss the ethical issues that arise when a developer uses AI to generate code that is then sold to a client.

INDICATIVE ANSWER / MARKING NOTE

Issues (any 4, 1 each): ownership/IP of generated code; potential plagiarism of training data; disclosure/transparency to the client; liability for defects; bias in output; data privacy of prompts. Reward balanced discussion of more than one viewpoint.

Q07**IDENTIFY****SECTION C · CASE STUDY****2 marks**

An organisation plans to embed an AI feature that recommends products to customers. Identify and describe one bias-related risk of this feature.

INDICATIVE ANSWER / MARKING NOTE

Identify (1): algorithmic/training-data bias. Describe (1): model may systematically disadvantage a group (e.g. recommend poorly for under-represented customers) because training data was unrepresentative.

Q08**EXPLAIN****SECTION C · CASE STUDY****3 marks**

Explain how a development team could responsibly integrate an AI coding assistant into their workflow while managing reliability concerns.

INDICATIVE ANSWER / MARKING NOTE

Any 3 × 1: human code review of all AI output; comprehensive testing (unit/integration); use AI for drafts not final logic; verify against requirements; document AI use; security scanning.

Q09**COMPARE****SECTION C · CASE STUDY****4 marks**

Compare AI-assisted development with traditional hand-coding in terms of speed, reliability, and developer skill.

INDICATIVE ANSWER / MARKING NOTE

Speed (1): AI faster for boilerplate. Reliability (1–2): hand-coding more predictable; AI needs verification. Skill (1): AI may erode/accelerate skill depending on use. Must explicitly compare, not list.

Q10**JUSTIFY****SECTION C · CASE STUDY****4 marks**

Ming asks an AI assistant to generate code for a payment module. Justify whether the organisation should deploy this code as-is.

INDICATIVE ANSWER / MARKING NOTE

Justification should conclude NO (or conditional): payment = high risk/security-sensitive; AI output unverified, possible vulnerabilities, compliance/PCI obligations, liability. Require review, testing, and audit before deployment. Reward a clear position supported by reasons.

Design tools for representing modules — IPO charts, pseudocode, object descriptions

2026 signal: Pseudocode is ~10–15% of every paper — effectively guaranteed.

Q01**IDENTIFY****SECTION A · MULTIPLE CHOICE****1 mark**

In an IPO chart, a running total that is updated each loop iteration is best classified as

- A. an input.
- B. a process.
- C. an output.
- D. a constant.

INDICATIVE ANSWER / MARKING NOTE

B. Accumulating a total is processing logic (input = raw values; output = displayed result).

Q02**STATE****SECTION B · SHORT ANSWER****1 mark**

State the purpose of pseudocode in the design stage of software development.

INDICATIVE ANSWER / MARKING NOTE

To represent the logic/steps of an algorithm in a language-independent, human-readable way before coding.

Q03**DESCRIBE****SECTION B · SHORT ANSWER****2 marks**

Describe what an IPO chart shows and one situation where it is useful.

INDICATIVE ANSWER / MARKING NOTE

Describe (1): maps the Inputs, Processing and Outputs of a module/solution. Use (1): clarifying data flow / requirements before designing the algorithm.

Q04**EXPLAIN****SECTION B · SHORT ANSWER****2 marks**

Write pseudocode that reads 10 integers and outputs the largest value.

INDICATIVE ANSWER / MARKING NOTE

Initialise largest to first/very small; loop 10 times reading n; IF n > largest THEN largest ← n; output largest. Reward correct initialisation, loop, comparison, output.

Q05

OUTLINE

SECTION B · SHORT ANSWER

3 marks

Trace the following pseudocode and state the final value of total. BEGIN total←0; FOR i←1 TO 4 DO total←total+i END FOR; PRINT total END

INDICATIVE ANSWER / MARKING NOTE

total = 1+2+3+4 = 10. (Award method marks for correct iteration values 1,3,6,10.)

Q06

DESCRIBE

SECTION B · SHORT ANSWER

2 marks

Describe the components you would include in an object description for a button object.

INDICATIVE ANSWER / MARKING NOTE

Object name/type; properties (e.g. caption, enabled, position); methods/events (e.g. onClick); behaviour/purpose. Any 2 components with detail.

Q07

EXPLAIN

SECTION C · CASE STUDY

3 marks

Write pseudocode for an algorithm that validates a password is at least 8 characters and contains a digit, looping until valid.

INDICATIVE ANSWER / MARKING NOTE

REPEAT input pwd; valid ← (length(pwd) ≥ 8) AND containsDigit(pwd); IF NOT valid THEN message UNTIL valid. Reward loop, both conditions, exit logic.

Q08

DESCRIBE

SECTION C · CASE STUDY

2 marks

Describe two advantages of producing pseudocode before writing program code.

INDICATIVE ANSWER / MARKING NOTE

Any 2 × 1: language-independent logic check; easier for others to follow/review; reduces coding errors; supports planning/communication.

Q09

EXPLAIN

SECTION C · CASE STUDY

3 marks

A team's pseudocode uses an inclusive boundary ($\leq$) where the requirement specifies 'under 18'. Explain the impact of this error and how a design tool helps catch it.

INDICATIVE ANSWER / MARKING NOTE

Impact (1-2): off-by-one/boundary error includes 18-year-olds incorrectly — wrong results. Design tool (1): reviewing pseudocode against requirements (or desk-checking/trace) exposes the boundary before coding.

Q10

COMPARE

SECTION C · CASE STUDY

4 marks

Compare pseudocode and an object description as design tools, including what each best represents.

INDICATIVE ANSWER / MARKING NOTE

Pseudocode (1-2): represents procedural logic/algorithms; sequence/selection/iteration. Object description (1-2): represents an object's properties, methods and behaviour. Must compare scope/purpose, not just define each.

U301 · KK06

UNIT 3 – OUTCOME 1 — PROGRAMMING

▲ VERY LIKELY

Characteristics of data sources — files, CSV and XML

2026 signal: CSV/XML named for 2026; XML caused 58% zero in 2025.

Q01

IDENTIFY

SECTION A · MULTIPLE CHOICE

1 mark

Which statement best describes XML compared with CSV?

- A. XML stores values separated by commas with no structure.
- B. XML uses nested tags to give data a hierarchical, self-describing structure.
- C. XML can only store numeric data.
- D. XML files are always smaller than equivalent CSV files.

INDICATIVE ANSWER / MARKING NOTE

B. XML is hierarchical and self-describing via tags (CSV is flat/delimited; D is false — XML is usually larger).

Q02

DEFINE

SECTION B · SHORT ANSWER

1 mark

Define the term 'delimiter' as used in a CSV file.

INDICATIVE ANSWER / MARKING NOTE

A character (e.g. a comma) that separates individual data values/fields within a record.

Q03

DESCRIBE

SECTION B · SHORT ANSWER

2 marks

Describe two characteristics of a CSV file.

INDICATIVE ANSWER / MARKING NOTE

Any 2 × 1: plain text; values separated by a delimiter (comma); one record per line; no inherent data types; flat/non-hierarchical; widely portable.

Q04

EXPLAIN

SECTION B · SHORT ANSWER

2 marks

Explain why XML is described as 'self-describing'.

INDICATIVE ANSWER / MARKING NOTE

Tags label the meaning of each element (e.g.), so the structure and meaning of the data travel with the data, without a separate schema being required to interpret field names.

Q05

COMPARE

SECTION B · SHORT ANSWER

3 marks

Compare CSV and XML as data sources for transferring structured data between systems.

INDICATIVE ANSWER / MARKING NOTE

CSV (1): compact, simple, flat, fast to parse, no hierarchy/types. XML (1): hierarchical, self-describing, supports nesting/attributes, larger/verbose. Comparison (1): choose CSV for simple tables, XML for nested/complex data. Must compare.

Q06

OUTLINE

SECTION B · SHORT ANSWER

2 marks

Write a small XML fragment that represents one student with a name and a year level.

INDICATIVE ANSWER / MARKING NOTE

...11 — reward well-formed nesting and a root/parent element with matching open/close tags.

Q07

EXPLAIN

SECTION C · CASE STUDY

3 marks

A solution imports a CSV where one row has a missing field. Explain why this can cause an error and how the program should handle it.

INDICATIVE ANSWER / MARKING NOTE

Cause (1-2): field count mismatch shifts/loses values, type conversion fails. Handling (1): validate field count, use existence/type checks, skip/flag malformed rows, error-handle the import.

Q08

DESCRIBE

SECTION C · CASE STUDY

2 marks

Describe one advantage and one disadvantage of using XML over CSV for storing an organisation's product catalogue.

INDICATIVE ANSWER / MARKING NOTE

Advantage (1): supports nested/structured data (categories, variants) and validation. Disadvantage (1): more verbose, larger files, slower/more complex to parse.

Q09

JUSTIFY

SECTION C · CASE STUDY

3 marks

A developer must exchange hierarchical configuration data between two systems. Justify whether CSV or XML is more appropriate.

INDICATIVE ANSWER / MARKING NOTE

Conclude XML: hierarchical/nested config needs structure CSV cannot represent; self-describing and validatable. Reward clear position + reasons tied to hierarchy.

Q10

EXPLAIN

SECTION C · CASE STUDY

3 marks

Explain how the choice of data source (CSV vs XML) affects the validation a developer must perform on import.

INDICATIVE ANSWER / MARKING NOTE

CSV (1-2): must check delimiter integrity, field count, types manually. XML (1-2): can validate well-formedness/schema, but must still check element presence and value ranges. Link choice to validation effort.

U301 · KK07

UNIT 3 - OUTCOME 1 — PROGRAMMING

▲ VERY LIKELY

Principles of object-oriented programming

2026 signal: OOP = 8 marks, avg only 3.5/8 — flagged the top priority.

Q01

IDENTIFY

SECTION A · MULTIPLE CHOICE

1 mark

Hiding an object's internal data and exposing only methods to access it is known as

- A. inheritance.
- B. polymorphism.
- C. encapsulation.
- D. abstraction.

INDICATIVE ANSWER / MARKING NOTE

C. Encapsulation bundles data with methods and restricts direct access.

Q02

DEFINE

SECTION B · SHORT ANSWER

2 marks

Define the terms 'class' and 'object' and state the relationship between them.

INDICATIVE ANSWER / MARKING NOTE

Class (1): a template/blueprint defining attributes and methods. Object (1): a specific instance of a class. Relationship: an object is created from a class.

Q03**DESCRIBE****SECTION B · SHORT ANSWER****2 marks**

Describe the principle of inheritance in object-oriented programming.

INDICATIVE ANSWER / MARKING NOTE

A subclass/child class acquires (inherits) the attributes and methods of a parent/superclass, and may extend or override them, promoting reuse.

Q04**EXPLAIN****SECTION B · SHORT ANSWER****2 marks**

Explain one benefit of encapsulation in a large software project.

INDICATIVE ANSWER / MARKING NOTE

Protects internal state from invalid external changes / reduces coupling, so internal implementation can change without breaking other code; improves maintainability and reliability.

Q05**DESCRIBE****SECTION B · SHORT ANSWER****2 marks**

Describe what polymorphism allows a programmer to do, with an example.

INDICATIVE ANSWER / MARKING NOTE

Allows the same method name/interface to behave differently across classes (e.g. draw() on Circle vs Square). Reward definition + valid example.

Q06**CONTRAST****SECTION B · SHORT ANSWER****3 marks**

Contrast a class and an object, and contrast an attribute and a method.

INDICATIVE ANSWER / MARKING NOTE

Class vs object (1-2): template vs instance. Attribute vs method (1): attribute = data/property of object; method = behaviour/operation. Must contrast each pair.

Q07**OUTLINE****SECTION C · CASE STUDY****2 marks**

An Account class has subclasses SavingsAccount and ChequeAccount. Outline how inheritance is used here.

INDICATIVE ANSWER / MARKING NOTE

Account holds shared attributes/methods (balance, deposit); the two subclasses inherit these and add/override specific behaviour (e.g. interest, overdraft).

Q08

EXPLAIN

SECTION C · CASE STUDY

3 marks

Explain how encapsulation and inheritance together improve the maintainability of an object-oriented solution.

INDICATIVE ANSWER / MARKING NOTE

Encapsulation (1-2): localises change behind interfaces. Inheritance (1-2): centralises shared code so a fix in the parent propagates. Together reduce duplication and limit ripple effects.

Q09

JUSTIFY

SECTION C · CASE STUDY

4 marks

A team debates using OOP versus procedural code for a banking application with many account types. Justify which paradigm is more suitable.

INDICATIVE ANSWER / MARKING NOTE

Conclude OOP: many related entities (account types) benefit from classes/inheritance/encapsulation for reuse, data protection, scalability. Reward position + at least two OOP principles applied to the scenario.

Q10

DESCRIBE

SECTION C · CASE STUDY

2 marks

Describe one disadvantage of using object-oriented programming for a very small, simple script.

INDICATIVE ANSWER / MARKING NOTE

Adds overhead/complexity (class structure) disproportionate to the task; can be slower to write/understand than a short procedural script.

U301-KK14

UNIT 3 – OUTCOME 1 — PROGRAMMING

▲ VERY LIKELY

Debugging and testing techniques

2026 signal: *Testing tables explicitly named as a 2026 revisit.*

Q01

IDENTIFY

SECTION A · MULTIPLE CHOICE

1 mark

A test case that uses a value exactly on the limit of an accepted range is testing

- A. typical data.
- B. boundary data.
- C. invalid data.
- D. erroneous logic.

INDICATIVE ANSWER / MARKING NOTE

B. Boundary (edge) data tests the limits of valid ranges.

Q02

STATE

SECTION B · SHORT ANSWER

1 mark

State the three categories of test data commonly used to test a solution.

INDICATIVE ANSWER / MARKING NOTE

Valid/typical (normal), boundary (edge), and invalid/erroneous data.

Q03

DESCRIBE

SECTION B · SHORT ANSWER

2 marks

Describe the columns you would include in a test table (test data plan).

INDICATIVE ANSWER / MARKING NOTE

Test no./case; input/test data; data type (valid/boundary/invalid); expected result; actual result; pass/fail. Award for naming purposeful columns.

Q04

EXPLAIN

SECTION B · SHORT ANSWER

2 marks

A field accepts ages 18–65. Explain why both 17 and 18 should appear in the test table.

INDICATIVE ANSWER / MARKING NOTE

17 (1): invalid/boundary-below should be rejected. 18 (1): valid boundary should be accepted; together they confirm the boundary is handled correctly (catches off-by-one).

Q05

OUTLINE

SECTION B · SHORT ANSWER

2 marks

Outline the difference between a syntax error and a logic error.

INDICATIVE ANSWER / MARKING NOTE

Syntax (1): breaks language rules; code won't compile/run. Logic (1): runs but produces wrong output; harder to detect.

Q06

DESCRIBE

SECTION B · SHORT ANSWER

2 marks

Describe two debugging techniques a programmer could use to locate a logic error.

INDICATIVE ANSWER / MARKING NOTE

Any 2 × 1: breakpoints/stepping; watch variables; print/trace statements; desk-checking; commenting out code; using a debugger to inspect state.

Q07

COMPARE

SECTION C · CASE STUDY

3 marks

Compare desk-checking and using a debugger as techniques for finding errors.

INDICATIVE ANSWER / MARKING NOTE

Desk-check (1): manual trace on paper, no execution, good early/conceptual. Debugger (1): runtime inspection, breakpoints, exact state. Comparison (1): desk-check catches logic in design; debugger catches runtime behaviour. Must compare.

Q08

EXPLAIN

SECTION C · CASE STUDY

3 marks

For a function that calculates a 10% discount on orders over \$100, design three test cases (valid, boundary, invalid) and state the expected output of each.

INDICATIVE ANSWER / MARKING NOTE

E.g. \$150 → \$15 discount (valid); \$100 → \$0 / boundary behaviour per spec; -\$50 or text → rejected (invalid). Reward one of each category with correct expected result.

Q09

EXPLAIN

SECTION C · CASE STUDY

3 marks

Explain why testing should use a documented test table rather than ad-hoc testing.

INDICATIVE ANSWER / MARKING NOTE

Any 3 × 1: repeatable/consistent; coverage of valid+boundary+invalid; evidence for evaluation/accountability; expected vs actual makes failures explicit; supports regression testing.

Q10

JUSTIFY

SECTION C · CASE STUDY

3 marks

A developer says testing only valid data is sufficient. Justify why this is inadequate.

INDICATIVE ANSWER / MARKING NOTE

Inadequate: valid-only ignores boundary (off-by-one) and invalid input (crashes/security). Robust solutions must reject bad input and handle edges. Reward clear stance + boundary AND invalid reasoning.

U302 · KK03

UNIT 3 – OUTCOME 2 — ANALYSIS & DESIGN

▲ VERY LIKELY

Project management — Gantt charts, dependencies, critical path, milestones

2026 signal: Gantt / critical-path annotations recur; ~40% scored zero in 2025.

Q01

IDENTIFY

SECTION A · MULTIPLE CHOICE

1 mark

On a Gantt chart, the longest sequence of dependent tasks that determines the minimum project duration is the

- A. milestone.
- B. slack time.
- C. critical path.
- D. dependency arrow.

INDICATIVE ANSWER / MARKING NOTE

C. The critical path determines the minimum completion time.

Q02

DEFINE

SECTION B · SHORT ANSWER

1 mark

Define the term 'milestone' in project management.

INDICATIVE ANSWER / MARKING NOTE

A significant point/event marking the completion of a phase or key deliverable; has zero duration.

Q03

DESCRIBE

SECTION B · SHORT ANSWER

2 marks

Describe what a dependency between two tasks means and how it is shown on a Gantt chart.

INDICATIVE ANSWER / MARKING NOTE

Dependency (1): one task cannot start (or finish) until another is complete. Shown (1): a linking arrow from predecessor to successor.

Q04

EXPLAIN

SECTION B · SHORT ANSWER

2 marks

Explain the significance of the critical path to a project manager.

INDICATIVE ANSWER / MARKING NOTE

Tasks on it have no slack; any delay delays the whole project, so the manager monitors/protects these tasks and allocates resources to them first.

Q05

IDENTIFY

SECTION B · SHORT ANSWER

2 marks

Tasks: A(2d)→C(3d); B(4d)→C. Identify the critical path and the project duration.

INDICATIVE ANSWER / MARKING NOTE

B→C = 7 days vs A→C = 5 days. Critical path = B→C; duration = 7 days.

Q06

OUTLINE

SECTION B · SHORT ANSWER

2 marks

Outline what 'slack' (float) is and why a non-critical task may have slack.

INDICATIVE ANSWER / MARKING NOTE

Slack (1): the time a task can be delayed without delaying the project. Reason (1): it runs in parallel with a longer dependent chain, so spare time exists.

Q07

EXPLAIN

SECTION C · CASE STUDY

3 marks

A developer falls two days behind on a task that is on the critical path. Explain the impact on the project and one way to recover.

INDICATIVE ANSWER / MARKING NOTE

Impact (1-2): whole project end date slips by ~2 days (no slack). Recovery (1): crash the task (add resources), fast-track parallelisable work, or re-scope.

Q08

DESCRIBE

SECTION C · CASE STUDY

2 marks

Describe two benefits of using a Gantt chart to manage a software development project.

INDICATIVE ANSWER / MARKING NOTE

Any 2 × 1: visualises schedule/timeline; shows dependencies; identifies critical path; tracks progress vs plan; aids resource allocation/communication.

Q09

EXPLAIN

SECTION C · CASE STUDY

3 marks

Explain how milestones and dependencies are used together to monitor whether a project is on track.

INDICATIVE ANSWER / MARKING NOTE

Dependencies (1–2): enforce task order and reveal flow-on delays. Milestones (1): checkpoints to compare planned vs actual completion, triggering corrective action if missed.

Q10

JUSTIFY

SECTION C · CASE STUDY

3 marks

A manager wants to shorten the project. Justify why adding resources to a non-critical task will NOT help.

INDICATIVE ANSWER / MARKING NOTE

Non-critical tasks have slack and are not the limiting chain; only shortening critical-path tasks reduces total duration. Reward correct reasoning about the critical path being the constraint.

U302 · KK04

UNIT 3 – OUTCOME 2 — ANALYSIS & DESIGN

▲ VERY LIKELY

Techniques for collecting data

2026 signal: Data-collection staple named for 2026; polarised 2025 results.

Q01

IDENTIFY

SECTION A · MULTIPLE CHOICE

1 mark

Which data-collection technique is MOST suitable for gathering in-depth views from a single key stakeholder?

- A. Survey.
- B. Interview.
- C. Observation.
- D. System log analysis.

INDICATIVE ANSWER / MARKING NOTE

B. Interviews suit deep, qualitative responses from individuals.

Q02**STATE****SECTION B · SHORT ANSWER****1 mark**

State one advantage of using a survey to collect requirements from a large user base.

INDICATIVE ANSWER / MARKING NOTE

Reaches many people quickly/cheaply and yields quantifiable data.

Q03**DESCRIBE****SECTION B · SHORT ANSWER****2 marks**

Describe the difference between an interview and a survey as data-collection techniques.

INDICATIVE ANSWER / MARKING NOTE

Interview (1): one-on-one, in-depth, qualitative, fewer people. Survey (1): standardised questions to many, mostly quantitative, less depth.

Q04**EXPLAIN****SECTION B · SHORT ANSWER****2 marks**

Explain one reason observation may collect more accurate data than a survey.

INDICATIVE ANSWER / MARKING NOTE

Observation records actual behaviour directly, avoiding self-report bias/inaccuracy where users misremember or misstate what they do.

Q05**OUTLINE****SECTION B · SHORT ANSWER****2 marks**

Outline two factors an analyst should consider when choosing a data-collection technique.

INDICATIVE ANSWER / MARKING NOTE

Any 2 × 1: number of participants; depth vs breadth needed; time/cost; nature of data (qual/quant); stakeholder availability; privacy.

Q06**CONTRAST****SECTION B · SHORT ANSWER****2 marks**

Contrast qualitative and quantitative data collected during analysis.

INDICATIVE ANSWER / MARKING NOTE

Qualitative (1): descriptive opinions/themes (interviews). Quantitative (1): numeric/measurable (survey counts). Must contrast.

Q07**DESCRIBE****SECTION C · CASE STUDY****2 marks**

An analyst will collect requirements from 300 staff and 3 managers. Describe a suitable technique for each group.

INDICATIVE ANSWER / MARKING NOTE

300 staff (1): survey/questionnaire for breadth. 3 managers (1): interviews for depth/strategic needs. Reward technique matched to group size/purpose.

Q08

EXPLAIN

SECTION C · CASE STUDY

3 marks

Explain how poorly designed survey questions could reduce the quality of collected requirements.

INDICATIVE ANSWER / MARKING NOTE

Any 3 × 1: leading/ambiguous questions bias answers; double-barrelled questions confuse; closed-only limits insight; poor sampling unrepresentative — leading to incorrect requirements.

Q09

JUSTIFY

SECTION C · CASE STUDY

3 marks

Justify the use of observation in addition to interviews when analysing an existing manual process.

INDICATIVE ANSWER / MARKING NOTE

Interviews capture stated process; observation reveals actual/undocumented steps and workarounds, validating/triangulating requirements. Reward triangulation reasoning.

Q10

DISCUSS

SECTION C · CASE STUDY

4 marks

Discuss the advantages and limitations of using two contrasting data-collection techniques when gathering requirements for a new app.

INDICATIVE ANSWER / MARKING NOTE

Pick two (e.g. survey + interview); give 2 strengths and 2 limitations across them and note how they complement each other (breadth + depth). Reward balanced discussion.

U302 · KK08

UNIT 3 – OUTCOME 2 — ANALYSIS & DESIGN

▲ VERY LIKELY

Depicting relationships — context diagrams, data flow diagrams, use case diagrams

2026 signal: DFD/use-case core but brutal — 81% zero on DFD, 52% on includes/extends.

Q01

IDENTIFY

SECTION A · MULTIPLE CHOICE

1 mark

In a data flow diagram, an open-ended rectangle (or rounded store symbol) represents a

- A. process.
- B. external entity.
- C. data store.
- D. data flow.

INDICATIVE ANSWER / MARKING NOTE

C. A data store holds data at rest.

Q02**DEFINE****SECTION B · SHORT ANSWER****1 mark**

Define the term 'external entity' in a context diagram.

INDICATIVE ANSWER / MARKING NOTE

A person, organisation or system outside the solution boundary that supplies or receives data.

Q03**DESCRIBE****SECTION B · SHORT ANSWER****2 marks**

Describe the difference between a context diagram and a level-0 data flow diagram.

INDICATIVE ANSWER / MARKING NOTE

Context (1): one process (the whole system) with external entities and net data flows. DFD level-0 (1): decomposes that into multiple processes, data stores and internal flows.

Q04**EXPLAIN****SECTION B · SHORT ANSWER****2 marks**

In a use case diagram, explain the difference between an 'include' and an 'extend' relationship.

INDICATIVE ANSWER / MARKING NOTE

Include (1): base use case always uses the included behaviour (mandatory sub-task). Extend (1): optional/conditional behaviour added under certain conditions.

Q05**IDENTIFY****SECTION B · SHORT ANSWER****2 marks**

For a login feature, identify one actor and one use case, and state their relationship.

INDICATIVE ANSWER / MARKING NOTE

Actor (1): User/Member. Use case (1): 'Log in'. Association: the actor initiates/participates in the use case (line connecting them).

Q06**OUTLINE****SECTION B · SHORT ANSWER****2 marks**

Outline the four key symbols used in a data flow diagram.

INDICATIVE ANSWER / MARKING NOTE

Process; external entity; data store; data flow (arrow). Award per correct symbol/role up to 2.

Q07**DESCRIBE****SECTION C · CASE STUDY****3 marks**

A library system: members borrow and return books; staff add new titles. Describe the actors and two use cases, including one include relationship.

INDICATIVE ANSWER / MARKING NOTE

Actors (1): Member, Staff. Use cases (1): Borrow book, Return book, Add title. Include (1): 'Borrow book' includes 'Verify membership' (always required).

Q08

EXPLAIN

SECTION C · CASE STUDY

3 marks

Explain why a context diagram is drawn before a detailed DFD, and what each contributes to analysis.

INDICATIVE ANSWER / MARKING NOTE

Context (1-2): fixes the system boundary and external interactions first, agreeing scope. DFD (1): then details internal processing/stores once the boundary is agreed; prevents scope confusion.

Q09

CONTRAST

SECTION C · CASE STUDY

3 marks

Contrast a use case diagram and a data flow diagram in terms of what each is designed to show.

INDICATIVE ANSWER / MARKING NOTE

Use case (1-2): functional requirements — who (actors) does what (use cases). DFD (1-2): how data moves/transforms between processes and stores. Must contrast purpose.

Q10

JUSTIFY

SECTION C · CASE STUDY

3 marks

An analyst omits data stores from a DFD. Justify why this is a significant error.

INDICATIVE ANSWER / MARKING NOTE

Data stores show where data persists; omitting them hides storage requirements, breaks data-flow balancing, and misrepresents the system, leading to design/build errors. Reward clear consequence reasoning.

U302 · KK10

UNIT 3 - OUTCOME 2 — ANALYSIS & DESIGN

▲ VERY LIKELY

Key legal requirements — privacy and copyright

2026 signal: Copyright & privacy legislation named for 2026.

Q01

IDENTIFY

SECTION A · MULTIPLE CHOICE

1 mark

Which Act primarily governs how Australian organisations handle individuals' personal information?

- A. Copyright Act 1968.
- B. Privacy Act 1988.
- C. Health Records Act 2001.
- D. Spam Act 2003.

INDICATIVE ANSWER / MARKING NOTE

B. The Privacy Act 1988 (and the Australian Privacy Principles) governs personal information handling.

Q02**STATE****SECTION B · SHORT ANSWER****1 mark**

State what the Copyright Act 1968 protects in the context of software.

INDICATIVE ANSWER / MARKING NOTE

The original expression of source code and related materials as a literary work, protecting against unauthorised copying/use.

Q03**DESCRIBE****SECTION B · SHORT ANSWER****2 marks**

Describe two obligations the Privacy Act 1988 places on an organisation that collects personal data.

INDICATIVE ANSWER / MARKING NOTE

Any 2 × 1: collect only what's necessary/with consent; notify purpose; keep secure; allow access/correction; use only for stated purpose; limit disclosure.

Q04**EXPLAIN****SECTION B · SHORT ANSWER****2 marks**

Explain one way a software solution can be designed to comply with the Privacy Act 1988.

INDICATIVE ANSWER / MARKING NOTE

E.g. encrypt stored personal data + access controls; collect minimum data; obtain consent; provide data access/correction features. Reward a concrete compliance measure tied to an obligation.

Q05**OUTLINE****SECTION B · SHORT ANSWER****2 marks**

Outline the difference between privacy and security of data.

INDICATIVE ANSWER / MARKING NOTE

Privacy (1): who may access/use personal data and for what purpose (rights). Security (1): technical/physical protection of data from unauthorised access/loss.

Q06**IDENTIFY****SECTION B · SHORT ANSWER****2 marks**

A developer copies a function from a tutorial website into a commercial product. Identify the legal issue and one way to address it.

INDICATIVE ANSWER / MARKING NOTE

Issue (1): potential copyright infringement of the author's code. Address (1): check/obtain a licence (e.g. permissive open-source) or write own implementation/attribute as required.

Q07

DESCRIBE

SECTION C · CASE STUDY

2 marks

Identify and describe one technique a developer could use to reduce a privacy risk when storing customer records.

INDICATIVE ANSWER / MARKING NOTE

Identify (1): encryption / access control / de-identification. Describe (1): how it limits unauthorised access to personal information (e.g. encrypt at rest so stolen data is unreadable).

Q08

EXPLAIN

SECTION C · CASE STUDY

3 marks

Explain the legal consequences for an organisation that fails to protect personal data it has collected.

INDICATIVE ANSWER / MARKING NOTE

Any 3 × 1: breach of Privacy Act/APPs; investigation/penalties by the regulator (OAIC); mandatory data-breach notification; reputational/financial damage; potential civil action.

Q09

DISCUSS

SECTION C · CASE STUDY

4 marks

Discuss the privacy and copyright issues a startup must consider before releasing a mobile app that uses third-party libraries and collects user location.

INDICATIVE ANSWER / MARKING NOTE

Copyright (1-2): library licences/attribution/permitted use. Privacy (1-2): consent for location, secure storage, purpose limitation, APP compliance. Reward both areas with applied detail.

Q10

JUSTIFY

SECTION C · CASE STUDY

3 marks

Justify why 'collect only what you need' is a sound design principle under privacy law.

INDICATIVE ANSWER / MARKING NOTE

Minimising collection reduces breach impact, aligns with APP data-minimisation, lowers compliance burden and user risk. Reward clear stance + link to legal obligation/risk reduction.

U302 · KK15

UNIT 3 – OUTCOME 2 — ANALYSIS & DESIGN

▲ VERY LIKELY

Characteristics of user experiences (UX)

2026 signal: UX & affordance named for 2026; up to 62% scored zero.

Q01**IDENTIFY****SECTION A · MULTIPLE CHOICE****1 mark**

A button styled to look raised and clickable is an example of

- A. consistency.
- B. affordance.
- C. redundancy.
- D. accessibility.

INDICATIVE ANSWER / MARKING NOTE

B. Affordance — the visual cue communicates how the element can be used.

Q02**DEFINE****SECTION B · SHORT ANSWER****1 mark**

Define the term 'affordance' in user experience design.

INDICATIVE ANSWER / MARKING NOTE

A property/cue of an element that signals how it can be used (e.g. a raised button affords pressing).

Q03**DESCRIBE****SECTION B · SHORT ANSWER****2 marks**

Describe two characteristics that contribute to a positive user experience.

INDICATIVE ANSWER / MARKING NOTE

Any 2 × 1: usability/ease of use; learnability; efficiency; accessibility; error tolerance; satisfaction/aesthetics; consistency.

Q04**EXPLAIN****SECTION B · SHORT ANSWER****2 marks**

Explain how good affordance reduces user error.

INDICATIVE ANSWER / MARKING NOTE

Clear cues make the correct action obvious, so users select intended controls without guessing, reducing mistaken or invalid interactions.

Q05**OUTLINE****SECTION B · SHORT ANSWER****2 marks**

Outline the difference between usability and accessibility.

INDICATIVE ANSWER / MARKING NOTE

Usability (1): how easily/efficiently typical users achieve goals. Accessibility (1): whether users with disabilities/diverse needs can use the solution.

Q06

DESCRIBE

SECTION B · SHORT ANSWER

2 marks

Describe one way an interface can provide useful feedback to the user.

INDICATIVE ANSWER / MARKING NOTE

E.g. confirmation message after submit, progress indicator, inline validation error, visual state change — keeps the user informed about the result of an action.

Q07

EXPLAIN

SECTION C · CASE STUDY

3 marks

A new app has many features but low user satisfaction. Explain three UX characteristics the team should evaluate.

INDICATIVE ANSWER / MARKING NOTE

Any 3 × 1 (with relevance): learnability, efficiency, error tolerance, consistency, feedback, accessibility — each linked to improving satisfaction.

Q08

DESCRIBE

SECTION C · CASE STUDY

2 marks

Describe how affordance and feedback work together when a user submits an online form.

INDICATIVE ANSWER / MARKING NOTE

Affordance (1): submit button looks pressable/active. Feedback (1): after pressing, a confirmation/error tells the user the outcome — closing the interaction loop.

Q09

DISCUSS

SECTION C · CASE STUDY

4 marks

Discuss how a developer can balance an attractive visual design with strong usability for a diverse user base.

INDICATIVE ANSWER / MARKING NOTE

Tension: aesthetics vs clarity/accessibility. Strategies (any 3–4): maintain affordance/consistency, sufficient contrast, accessible controls, test with real users, avoid form-over-function. Reward balanced discussion.

Q10

JUSTIFY

SECTION C · CASE STUDY

3 marks

Justify why removing all button borders and labels for a 'clean' look could harm UX.

INDICATIVE ANSWER / MARKING NOTE

Removes affordance/feedback cues; users can't tell what is interactive, raising errors and cognitive load — aesthetics shouldn't override discoverability. Reward stance + affordance reasoning.

U302 · KK16

UNIT 3 — OUTCOME 2 — ANALYSIS & DESIGN

▲ VERY LIKELY

Design principles influencing appearance and functionality

Q01

IDENTIFY

SECTION A · MULTIPLE CHOICE

1 mark

Using the same colours, fonts and button styles across all screens is an example of the design principle of

- A. alignment.
- B. contrast.
- C. consistency.
- D. proximity.

INDICATIVE ANSWER / MARKING NOTE

C. Consistency.

Q02

STATE

SECTION B · SHORT ANSWER

1 mark

State one appearance-related design principle and one functionality-related design principle.

INDICATIVE ANSWER / MARKING NOTE

Appearance (1): alignment/contrast/balance/colour. Functionality (1): navigation/consistency/accessibility/error prevention. Reward one valid each.

Q03

DESCRIBE

SECTION B · SHORT ANSWER

2 marks

Describe the design principle of contrast and one benefit it provides.

INDICATIVE ANSWER / MARKING NOTE

Contrast (1): using difference (colour/size/weight) to distinguish elements. Benefit (1): draws attention to key items / improves readability and hierarchy.

Q04

EXPLAIN

SECTION B · SHORT ANSWER

2 marks

Explain how the principle of proximity helps users understand a form.

INDICATIVE ANSWER / MARKING NOTE

Grouping related fields/labels close together signals they belong together, so users parse structure faster and complete the form with fewer errors.

Q05

OUTLINE

SECTION B · SHORT ANSWER

2 marks

Outline what 'user-centred design' means.

INDICATIVE ANSWER / MARKING NOTE

Designing around real users' needs/goals/abilities, involving them through analysis, prototyping and testing across the process.

Q06

DESCRIBE

SECTION B · SHORT ANSWER

2 marks

Describe how the principle of alignment improves the appearance of an interface.

INDICATIVE ANSWER / MARKING NOTE

Aligning elements to a grid/edges creates order and visual connection, making the layout look professional and easier to scan.

Q07

EXPLAIN

SECTION C · CASE STUDY

3 marks

Explain how three design principles could be applied to improve a cluttered dashboard.

INDICATIVE ANSWER / MARKING NOTE

Any 3 × 1 applied: proximity (group metrics), contrast (highlight key KPIs), alignment/white space (order), consistency (uniform components).

Q08

CONTRAST

SECTION C · CASE STUDY

3 marks

Contrast design principles that affect appearance with those that affect functionality, giving an example of each.

INDICATIVE ANSWER / MARKING NOTE

Appearance (1-2): contrast/balance/colour — look. Functionality (1-2): navigation/consistency/accessibility — behaviour/usability. Must contrast with examples.

Q09

DISCUSS

SECTION C · CASE STUDY

4 marks

Discuss how applying design principles supports both accessibility and a strong brand identity.

INDICATIVE ANSWER / MARKING NOTE

Accessibility (1-2): contrast, consistency, clear navigation aid diverse users. Brand (1-2): consistent colour/typography builds identity. Note possible tension and resolution. Reward balance.

Q10

JUSTIFY

SECTION C · CASE STUDY

3 marks

A client wants a unique font on every screen. Justify advice against this using design principles.

INDICATIVE ANSWER / MARKING NOTE

Breaks consistency, harms readability/recognition and increases cognitive load; recommend a limited, consistent type system. Reward stance + consistency/usability reasoning.

U401 · KK04

UNIT 4 – OUTCOME 1 — DEVELOPMENT & EVALUATION

▲ VERY LIKELY

Innovative approaches to software development

2026 signal: Innovative development methods named as a 2026 emphasis.

Q01**IDENTIFY****SECTION A · MULTIPLE CHOICE****1 mark**

An approach that delivers software in short, repeated cycles with frequent stakeholder feedback is best described as

- A. waterfall.
- B. agile/iterative.
- C. big-bang.
- D. ad-hoc.

INDICATIVE ANSWER / MARKING NOTE

B. Agile/iterative development.

Q02**DEFINE****SECTION B · SHORT ANSWER****1 mark**

Define the term 'iterative development'.

INDICATIVE ANSWER / MARKING NOTE

Building the solution in repeated cycles, each adding/refining functionality with review before the next cycle.

Q03**DESCRIBE****SECTION B · SHORT ANSWER****2 marks**

Describe two characteristics of an agile approach to software development.

INDICATIVE ANSWER / MARKING NOTE

Any 2 × 1: short iterations/sprints; frequent stakeholder collaboration; embracing changing requirements; incremental delivery; continuous testing/feedback.

Q04**EXPLAIN****SECTION B · SHORT ANSWER****2 marks**

Explain one benefit of using rapid prototyping during development.

INDICATIVE ANSWER / MARKING NOTE

Lets users see/try an early version and give feedback, so requirements/usability issues are found and fixed early, reducing costly late changes.

Q05**CONTRAST****SECTION B · SHORT ANSWER****3 marks**

Contrast an iterative (agile) approach with a sequential (waterfall) approach.

INDICATIVE ANSWER / MARKING NOTE

Iterative (1-2): cyclic, adaptive, frequent feedback, change-tolerant. Waterfall (1-2): linear phases, plan-heavy, change-resistant. Must contrast process and response to change.

Q06

OUTLINE

SECTION B · SHORT ANSWER

2 marks

Outline two ways emerging technologies can drive innovative approaches to development.

INDICATIVE ANSWER / MARKING NOTE

Any 2 × 1: AI code/test generation; cloud/DevOps continuous delivery; low-code platforms; automated testing pipelines; collaborative/remote tooling.

Q07

EXPLAIN

SECTION C · CASE STUDY

3 marks

A startup faces rapidly changing client requirements. Explain why an iterative approach suits this situation.

INDICATIVE ANSWER / MARKING NOTE

Any 3 × 1: short cycles absorb change cheaply; frequent feedback realigns priorities; early increments deliver value; reduces risk of building the wrong thing.

Q08

DESCRIBE

SECTION C · CASE STUDY

2 marks

Describe one risk of adopting an innovative/agile method for a team used to fixed, up-front planning.

INDICATIVE ANSWER / MARKING NOTE

E.g. scope can drift without discipline; harder to fix budget/deadline; requires cultural change/skills — may cause confusion or rework if poorly managed.

Q09

DISCUSS

SECTION C · CASE STUDY

4 marks

Discuss the advantages and disadvantages of incorporating AI-driven automation into a development pipeline.

INDICATIVE ANSWER / MARKING NOTE

Advantages (1–2): speed, automated testing/deployment, fewer manual errors. Disadvantages (1–2): reliability/trust, skill/setup cost, over-reliance, opacity. Reward balanced discussion.

Q10

JUSTIFY

SECTION C · CASE STUDY

4 marks

Justify the choice of an iterative approach over a waterfall approach for an app with uncertain, evolving requirements.

INDICATIVE ANSWER / MARKING NOTE

Conclude iterative: uncertainty + change favour short feedback cycles, early validation, adaptability; waterfall's fixed up-front spec is risky here. Reward clear position + 2+ reasons applied.

U402 · KK02

UNIT 4 – OUTCOME 2 — CYBER SECURITY

▲ VERY LIKELY

Advantages and disadvantages of developing software in-house vs externally

2026 signal: In-house vs external dev — only 1% scored full; due for retest.

Q01

IDENTIFY

SECTION A · MULTIPLE CHOICE

1 mark

One advantage of developing software in-house rather than outsourcing is

- A. lower up-front recruitment cost.
- B. greater control over the solution and data.
- C. access to a wider external skill pool.
- D. faster delivery for an organisation with no developers.

INDICATIVE ANSWER / MARKING NOTE

B. In-house gives direct control over the solution and sensitive data.

Q02

STATE

SECTION B · SHORT ANSWER

1 mark

State one disadvantage of developing software in-house.

INDICATIVE ANSWER / MARKING NOTE

Higher staffing/skills cost, slower if expertise is lacking, or limited specialist knowledge. (Any one.)

Q03

DESCRIBE

SECTION B · SHORT ANSWER

2 marks

Describe two advantages of developing software in-house.

INDICATIVE ANSWER / MARKING NOTE

Any 2 × 1: control over data/IP; alignment with business knowledge; easier ongoing maintenance; better security oversight; tailored solution.

Q04

DESCRIBE

SECTION B · SHORT ANSWER

2 marks

Describe two advantages of outsourcing (external) development.

INDICATIVE ANSWER / MARKING NOTE

Any 2 × 1: access to specialist skills; potentially faster/cheaper; scalable capacity; frees internal staff; vendor accountability.

Q05

EXPLAIN

SECTION B · SHORT ANSWER

2 marks

Explain one security risk that arises specifically when development is outsourced externally.

INDICATIVE ANSWER / MARKING NOTE

Sharing source/data with a third party increases exposure (data leakage, supply-chain compromise, less direct control over the vendor's security practices).

Q06

OUTLINE

SECTION B · SHORT ANSWER

2 marks

Outline two factors an organisation should weigh when deciding between in-house and external development.

INDICATIVE ANSWER / MARKING NOTE

Any 2 × 1: available internal skills; cost/budget; timeframe; sensitivity of data; need for ongoing control/maintenance; IP ownership.

Q07

COMPARE

SECTION C · CASE STUDY

4 marks

Compare in-house and external software development in terms of cost, control, and security.

INDICATIVE ANSWER / MARKING NOTE

Cost (1): in-house ongoing salary vs external project fee. Control (1-2): in-house higher; external less. Security (1-2): in-house contained; external adds third-party exposure. Must compare each, not list.

Q08

EXPLAIN

SECTION C · CASE STUDY

3 marks

A bank holding sensitive customer data must choose a development model. Explain why in-house may be preferred.

INDICATIVE ANSWER / MARKING NOTE

Any 3 × 1: tighter control of sensitive data; reduced third-party exposure; regulatory/compliance assurance; deep domain knowledge; long-term maintenance control.

Q09

DISCUSS

SECTION C · CASE STUDY

4 marks

Discuss the trade-offs a small business faces in outsourcing development of its first app.

INDICATIVE ANSWER / MARKING NOTE

Pros (1-2): skills, speed, lower up-front staffing. Cons (1-2): less control, data exposure, ongoing dependency/cost, IP terms. Reward balanced, applied discussion.

Q10

JUSTIFY

SECTION C · CASE STUDY

4 marks

Justify a recommendation for a startup with no developers and a tight deadline to outsource its software.

INDICATIVE ANSWER / MARKING NOTE

Conclude outsource: no internal capability + deadline → external skills/speed outweigh control loss, provided contracts cover IP/security/maintenance. Reward clear stance + reasons + risk mitigation.

U402 · KK03

UNIT 4 – OUTCOME 2 — CYBER SECURITY

▲ VERY LIKELY

Vulnerabilities within insecure development environments (incl. combined dev/test/production)

2026 signal: Combined dev environments named core; only 9% scored full marks.

Q01

IDENTIFY

SECTION A · MULTIPLE CHOICE

1 mark

A key risk of combining development, testing and production in a single environment is that

- A. code runs faster.
- B. untested or insecure code can directly affect live data.
- C. version control becomes unnecessary.
- D. developers cannot access the database.

INDICATIVE ANSWER / MARKING NOTE

B. Without separation, defective/insecure code and test actions can damage or expose live/production data.

Q02

DEFINE

SECTION B · SHORT ANSWER

1 mark

Define the term 'vulnerability' in a development environment.

INDICATIVE ANSWER / MARKING NOTE

A weakness in a system, practice or configuration that could be exploited to compromise security.

Q03

DESCRIBE

SECTION B · SHORT ANSWER

2 marks

Describe two vulnerabilities that can exist in an insecure development environment.

INDICATIVE ANSWER / MARKING NOTE

Any 2 × 1: unpatched software; combined dev/test/prod; poor access management; insecure APIs; weak code review; use of real data in testing; insider threats.

Q04

EXPLAIN

SECTION B · SHORT ANSWER

2 marks

Explain why using real customer data in a testing environment is a vulnerability.

INDICATIVE ANSWER / MARKING NOTE

Test environments are usually less secured/monitored; exposing real personal data there risks breach and privacy non-compliance if accessed/leaked.

Q05

OUTLINE

SECTION B · SHORT ANSWER

2 marks

Outline the purpose of separating development, testing and production environments.

INDICATIVE ANSWER / MARKING NOTE

Isolates unverified code/changes from live data and users, so defects/security issues are contained before release — protecting production integrity.

Q06

DESCRIBE

SECTION B · SHORT ANSWER

2 marks

Describe how unpatched software becomes a vulnerability during development.

INDICATIVE ANSWER / MARKING NOTE

Known exploits in outdated tools/libraries remain open; attackers can leverage published vulnerabilities to compromise the environment or injected dependencies.

Q07

EXPLAIN

SECTION C · CASE STUDY

3 marks

Explain three risks introduced by combining development, testing and production in one environment.

INDICATIVE ANSWER / MARKING NOTE

Any 3 × 1: untested code corrupts live data; test/debug exposure of production data; broad access increases insider/attack surface; outage of live system during dev/test; no clean rollback.

Q08

DESCRIBE

SECTION C · CASE STUDY

2 marks

Describe one control that mitigates the risk of combined environments.

INDICATIVE ANSWER / MARKING NOTE

E.g. separate isolated environments + role-based access + use de-identified/synthetic test data + controlled deployment pipeline. Reward control linked to separation.

Q09

DISCUSS

SECTION C · CASE STUDY

4 marks

Discuss why poor identity and access management is a serious vulnerability in a development environment.

INDICATIVE ANSWER / MARKING NOTE

Excessive/standing privileges, shared accounts, no least-privilege → wider attack surface, harder accountability/audit, easier insider misuse and lateral movement. Reward 3–4 linked points.

Q10

JUSTIFY

SECTION C · CASE STUDY

3 marks

Justify the cost of maintaining separate environments to management focused on saving money.

INDICATIVE ANSWER / MARKING NOTE

Separation prevents costly production breaches/outages and data exposure; the breach/compliance cost far exceeds environment cost. Reward stance + risk/cost reasoning.

U402 · KK04

UNIT 4 – OUTCOME 2 — CYBER SECURITY

▲ VERY LIKELY

Security controls to protect development practices and data

Q01

IDENTIFY

SECTION A · MULTIPLE CHOICE

1 mark

Encrypting data so it is unreadable without a key is an example of a

- A. physical control.
- B. technical (software) control.
- C. administrative control.
- D. detective audit.

INDICATIVE ANSWER / MARKING NOTE

B. Encryption is a technical/software control.

Q02

STATE

SECTION B · SHORT ANSWER

1 mark

State one physical and one software (technical) security control.

INDICATIVE ANSWER / MARKING NOTE

Physical (1): locked server room/biometric door. Software (1): encryption/firewall/access control.
Reward one valid each.

Q03

DESCRIBE

SECTION B · SHORT ANSWER

2 marks

Describe two software security controls that protect data within an application.

INDICATIVE ANSWER / MARKING NOTE

Any 2 × 1: encryption; authentication; access control/authorisation; input validation; logging/auditing; firewalls.

Q04

EXPLAIN

SECTION B · SHORT ANSWER

2 marks

Explain how the principle of least privilege protects an organisation's data.

INDICATIVE ANSWER / MARKING NOTE

Users/processes get only the access needed for their role, limiting damage from compromised accounts/insiders and shrinking the attack surface.

Q05

OUTLINE

SECTION B · SHORT ANSWER

2 marks

Outline the difference between authentication and authorisation.

INDICATIVE ANSWER / MARKING NOTE

Authentication (1): verifying who a user is (login/credentials). Authorisation (1): what an authenticated user is allowed to do/access.

Q06

DESCRIBE

SECTION B · SHORT ANSWER

2 marks

Describe how multi-factor authentication improves security.

INDICATIVE ANSWER / MARKING NOTE

Requires two+ independent factors (e.g. password + device code), so a stolen password alone cannot grant access — reducing account compromise.

Q07

EXPLAIN

SECTION C · CASE STUDY

3 marks

For a web app storing payment data, explain three security controls the team should implement.

INDICATIVE ANSWER / MARKING NOTE

Any 3 × 1 applied: encryption at rest/in transit; strong authentication/MFA; least-privilege access; input validation; logging/monitoring; patching.

Q08

DESCRIBE

SECTION C · CASE STUDY

2 marks

Identify and describe one administrative control that supports secure development.

INDICATIVE ANSWER / MARKING NOTE

Identify (1): security policy / developer training / code-review process / access-review procedure.
Describe (1): how it reduces risk (e.g. mandatory peer review catches insecure code before release).

Q09

DISCUSS

SECTION C · CASE STUDY

4 marks

Discuss how a combination of physical, software and administrative controls provides defence in depth.

INDICATIVE ANSWER / MARKING NOTE

Defence in depth (1): layered controls so one failure doesn't breach the system. Give one of each layer (1×3) with how they complement. Reward layered reasoning.

Q10

JUSTIFY

SECTION C · CASE STUDY

3 marks

Justify implementing encryption even when access controls are already in place.

INDICATIVE ANSWER / MARKING NOTE

Access controls can be bypassed/misconfigured or data stolen physically; encryption keeps data unreadable as a second layer — defence in depth. Reward stance + layering reasoning.

U402 · KK05

UNIT 4 — OUTCOME 2 — CYBER SECURITY

▲ VERY LIKELY

Threat modelling principles

2026 signal: Threat modelling named as a 2026 revisit.

Q01**IDENTIFY****SECTION A · MULTIPLE CHOICE****1 mark**

Which of the following is a principle of threat modelling?

- A. Releasing software as early as possible.
- B. Identifying assets, threats and vulnerabilities, then confirming mitigations.
- C. Avoiding documentation to save time.
- D. Granting all developers admin access.

INDICATIVE ANSWER / MARKING NOTE

B. Threat modelling identifies what to protect, the threats/vulnerabilities, and verifies mitigations.

Q02**DEFINE****SECTION B · SHORT ANSWER****1 mark**

Define the term 'threat modelling'.

INDICATIVE ANSWER / MARKING NOTE

A structured process of identifying assets, potential threats and vulnerabilities, and planning/verifying mitigations during design.

Q03**OUTLINE****SECTION B · SHORT ANSWER****2 marks**

Outline two questions a team asks when threat modelling a new system.

INDICATIVE ANSWER / MARKING NOTE

Any 2 × 1: What are we building / what assets exist? What can go wrong (threats)? What are we doing about it (mitigations)? Did we do a good enough job (verify)?

Q04**DESCRIBE****SECTION B · SHORT ANSWER****2 marks**

Describe the difference between a threat and a vulnerability.

INDICATIVE ANSWER / MARKING NOTE

Threat (1): a potential cause of harm (attacker/event). Vulnerability (1): a weakness the threat could exploit. (Risk = likelihood × impact of a threat exploiting a vulnerability.)

Q05**EXPLAIN****SECTION B · SHORT ANSWER****2 marks**

Explain why threat modelling is best done early in the design stage.

INDICATIVE ANSWER / MARKING NOTE

Designing in mitigations is cheaper and more effective than retrofitting after build; early modelling prevents architectural weaknesses and reduces costly late fixes.

Q06

IDENTIFY

SECTION B · SHORT ANSWER

2 marks

For an online banking app, identify one asset and one threat to that asset.

INDICATIVE ANSWER / MARKING NOTE

Asset (1): customer account data/funds. Threat (1): credential theft/phishing, SQL injection, insider misuse. Reward asset + matched threat.

Q07

DESCRIBE

SECTION C · CASE STUDY

3 marks

Describe the steps of a simple threat modelling process for a new web application.

INDICATIVE ANSWER / MARKING NOTE

1 identify assets/scope; 2 identify threats (e.g. STRIDE-style); 3 identify vulnerabilities; 4 plan mitigations; 5 confirm/verify mitigations. Award for ordered, sensible steps.

Q08

EXPLAIN

SECTION C · CASE STUDY

3 marks

Explain how threat modelling informs the choice of security controls in a solution.

INDICATIVE ANSWER / MARKING NOTE

Identified threats/assets are prioritised by risk; controls are selected to mitigate the highest-risk threats (e.g. injection → input validation), ensuring controls are targeted not arbitrary.

Q09

DISCUSS

SECTION C · CASE STUDY

4 marks

Discuss the benefits and limitations of threat modelling as a security practice.

INDICATIVE ANSWER / MARKING NOTE

Benefits (1-2): proactive, prioritised, early design improvement, shared understanding. Limitations (1-2): only as good as inputs/expertise, can miss novel threats, time cost, needs updating. Reward balance.

Q10

JUSTIFY

SECTION C · CASE STUDY

3 marks

Justify spending time on threat modelling for a project with a tight deadline.

INDICATIVE ANSWER / MARKING NOTE

Early identification of high-risk threats prevents far costlier breaches/rework later; targeted modelling need not be exhaustive. Reward stance + cost/risk trade-off reasoning.

U402 · KK07

UNIT 4 – OUTCOME 2 — CYBER SECURITY

▲ VERY LIKELY

Key legislation affecting organisations and data

2026 signal: Privacy / copyright legislation named for 2026.

Q01

IDENTIFY

SECTION A · MULTIPLE CHOICE

1 mark

An organisation suffers a data breach likely to cause serious harm. Under Australian law it must

- A. do nothing unless customers complain.
- B. notify affected individuals and the OAIC.
- C. only notify if more than 1000 records are involved.
- D. delete all remaining data immediately.

INDICATIVE ANSWER / MARKING NOTE

B. The Notifiable Data Breaches scheme requires notifying affected individuals and the OAIC.

Q02

STATE

SECTION B · SHORT ANSWER

1 mark

State the purpose of the Australian Privacy Principles (APPs).

INDICATIVE ANSWER / MARKING NOTE

They set out how organisations must collect, use, store, secure and disclose personal information.

Q03

DESCRIBE

SECTION B · SHORT ANSWER

2 marks

Describe two requirements the Privacy Act 1988 places on data handling.

INDICATIVE ANSWER / MARKING NOTE

Any 2 × 1: collect lawfully/with consent; use only for stated purpose; secure storage; allow access/correction; limit disclosure; notify breaches.

Q04

EXPLAIN

SECTION B · SHORT ANSWER

2 marks

Explain how the Copyright Act 1968 applies to source code written by an employee.

INDICATIVE ANSWER / MARKING NOTE

Original code is a protected literary work; copyright typically vests in the employer for work made in employment, restricting others from copying/using it without permission.

Q05

OUTLINE

SECTION B · SHORT ANSWER

2 marks

Outline what the Notifiable Data Breaches scheme requires.

INDICATIVE ANSWER / MARKING NOTE

Where a breach is likely to cause serious harm, the organisation must promptly notify affected individuals and the regulator (OAIC) and describe the breach and recommended steps.

Q06

IDENTIFY

SECTION B · SHORT ANSWER

2 marks

Identify two pieces of legislation relevant to an organisation that stores and processes personal data, and state one obligation from each.

INDICATIVE ANSWER / MARKING NOTE

Privacy Act 1988 (1): secure/limit personal data. Copyright Act 1968 (1): don't copy protected works; or Health Records Act for health data. Reward two acts + an obligation each.

Q07

EXPLAIN

SECTION C · CASE STUDY

3 marks

Explain the consequences for an organisation that breaches privacy legislation.

INDICATIVE ANSWER / MARKING NOTE

Any 3 × 1: regulator investigation; civil penalties/fines; mandatory breach notification; remediation orders; reputational and financial harm; loss of customer trust.

Q08

DESCRIBE

SECTION C · CASE STUDY

2 marks

Describe how legislation influences design decisions in a data-handling solution.

INDICATIVE ANSWER / MARKING NOTE

Drives requirements such as encryption, consent capture, access/correction features, data minimisation, audit logging and breach-response capability — compliance becomes a design constraint.

Q09

DISCUSS

SECTION C · CASE STUDY

4 marks

Discuss the obligations an organisation has under privacy and copyright law when building software that reuses third-party code and stores user data.

INDICATIVE ANSWER / MARKING NOTE

Copyright (1-2): licence/permission/attribution for reused code. Privacy (1-2): consent, security, purpose limitation, breach notification. Reward both with applied detail.

Q10

JUSTIFY

SECTION C · CASE STUDY

3 marks

Justify why an organisation should maintain audit logs of access to personal data.

INDICATIVE ANSWER / MARKING NOTE

Supports privacy compliance/accountability, breach investigation/notification, and deters/detects misuse; evidences due diligence to the regulator. Reward stance + legal/security reasoning.

U402 · KK08

UNIT 4 - OUTCOME 2 — CYBER SECURITY

▲ VERY LIKELY

Ethical issues arising when developing software

Q01

IDENTIFY

SECTION A · MULTIPLE CHOICE

1 mark

A developer knowingly ships software with a privacy flaw to meet a deadline. This is primarily

- A. a legal issue only.
- B. an ethical issue concerning responsibility to users.
- C. a performance optimisation.
- D. acceptable if undocumented.

INDICATIVE ANSWER / MARKING NOTE

B. It raises an ethical responsibility to users (and may also be legal), centred on duty of care.

Q02

DEFINE

SECTION B · SHORT ANSWER

1 mark

Define the difference between a legal and an ethical obligation.

INDICATIVE ANSWER / MARKING NOTE

Legal: required by law with penalties. Ethical: a moral expectation of right conduct that may exceed what the law strictly requires.

Q03

DESCRIBE

SECTION B · SHORT ANSWER

2 marks

Describe two ethical issues that can arise when developing software that uses AI.

INDICATIVE ANSWER / MARKING NOTE

Any 2 × 1: bias/discrimination in outputs; lack of transparency; privacy of training/user data; accountability for harm; job displacement; misuse.

Q04

EXPLAIN

SECTION B · SHORT ANSWER

2 marks

Explain why bias in an AI model is an ethical concern.

INDICATIVE ANSWER / MARKING NOTE

Biased training data can lead the system to disadvantage particular groups unfairly, causing discrimination and harm while appearing objective — raising fairness/accountability concerns.

Q05

OUTLINE

SECTION B · SHORT ANSWER

2 marks

Outline two responsibilities a developer has towards end users.

INDICATIVE ANSWER / MARKING NOTE

Any 2 × 1: protect privacy/security of data; deliver reliable/safe software; be transparent/honest; avoid harm; accessibility/fairness.

Q06

DESCRIBE

SECTION B · SHORT ANSWER

2 marks

Describe one ethical issue related to collecting more user data than a feature requires.

INDICATIVE ANSWER / MARKING NOTE

Over-collection breaches data-minimisation/consent expectations, increases breach harm and surveillance risk, and treats users' data without due respect — an ethical (and legal) concern.

Q07

DISCUSS

SECTION C · CASE STUDY

4 marks

An organisation deploys an AI hiring tool. Discuss the ethical issues it should address before use.

INDICATIVE ANSWER / MARKING NOTE

Any 4 × 1: bias/fairness, transparency/explainability, accountability for decisions, candidate privacy/consent, human oversight, testing for discrimination. Reward more than one viewpoint.

Q08

EXPLAIN

SECTION C · CASE STUDY

3 marks

Explain how an organisation can act ethically when an AI feature could produce biased results.

INDICATIVE ANSWER / MARKING NOTE

Any 3 × 1: audit/test for bias on diverse data; keep human-in-the-loop; be transparent to users; allow appeal/correction; document and monitor; restrict scope.

Q09

CONTRAST

SECTION C · CASE STUDY

3 marks

Contrast acting legally with acting ethically when developing software that tracks user location.

INDICATIVE ANSWER / MARKING NOTE

Legal (1–2): meet Privacy Act/consent minimums. Ethical (1–2): may go further — only track when truly needed, clear opt-out, honest disclosure even where law is silent. Must contrast.

Q10

JUSTIFY

SECTION C · CASE STUDY

4 marks

A client asks a developer to add hidden data collection users won't notice. Justify the developer's ethical response.

INDICATIVE ANSWER / MARKING NOTE

Conclude refuse/disclose: hidden collection breaches transparency/consent and likely privacy law; ethical duty to users overrides client request; propose transparent, consented alternative. Reward stance + ethical (and legal) reasoning + alternative.